Chapter 5

Boundary Strength Scaling

Abstract

Prosodic grouping reflects syntactic structure. This chapter presents evidence that utterances are structured hierarchically by imposing boundaries of various ranks, reflecting syntactic levels of embedding, as proposed in chapter 2. Prosodic boundaries are scaled relative to boundaries produced earlier in the utterance. This phenomenon that will be called 'Boundary Strength Scaling' in analogy to pitch-accent scaling, which similarly depends on the hierarchical prosodic organization of an utterance.

5.1 Scaling the Strength of Boundaries

This chapter reports on an experiment testing the boundary ranks in coordinate structures assigned by the algorithm presented in chapter 2. The generalization about boundary ranks proposed relates the strength of boundaries to syntactic levels of embedding. The experimental evidence shows that the model is better than a competing model from the processing literature that assigns boundaries more locally only looking at the two adjacent syntactic constituents (Watson and Gibson, 2004a,b) .

A closer look at the data reveals that speakers scale boundaries relative to boundaries produced earlier in an utterance. I propose that the implementation of prosodic structure allows for 'Boundary Strength Scaling'. Boundary Strength scaling makes global syntactic influences on prosody compatible with incremental structure building in on-line production.

5.2 Global Syntactic Effects

Syntactic, phonological, and processing factors influence prosodic phrasing. Early accounts of phrasing in the processing literature (Cooper and Paccia-Cooper, 1980, Grosjean and Collins, 1979) explain effects on prosody in reference to syntactic structure. Gee and Grosjean (1983), Ferreira (1988, 1993) map syntactic structure to a prosodic structure, allowing for certain mismatches, e.g. in order to account for the phrasing of function words.

Watson and Gibson (2004a,b) argue for a local processing-based model that uses only the length of the constituent to the left and right of a boundary (LRB-Theory) to predict its strength. The LRB thus greatly simplifies the mapping between syntax and prosody, and is compatible with incremental structure building since only local information is needed.

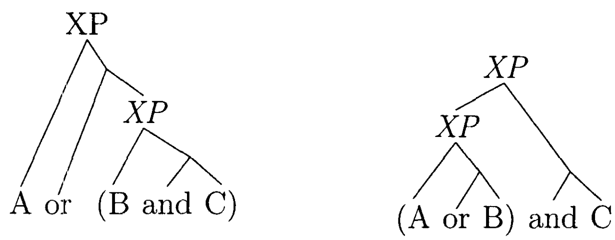
This section presents new evidence for a more syntactic approach to prosodic phrasing. In particular, it presents evidence for the boundary ranks assigned by the algorithm in chapter 2.

5.2.1 Coordinate Structures as a Test Case

In coordinate structures, different bracketings are coded by parentheses in orthography, and by prosody in speech:

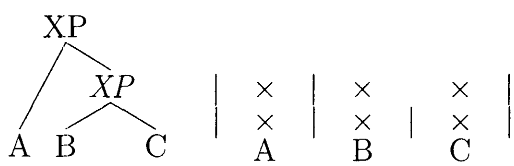
(1) Coordinate Structures

a. Right-Branching b. Left-Branching



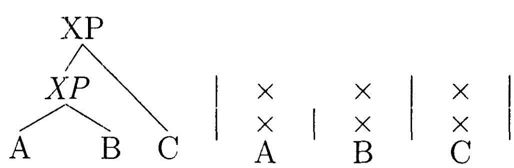
Prosodic phrasing reflects the bracketing of syntactic structure. In (1a), the boundary following A is stronger than the boundary following B:

(2)



In (1b), the reverse is true:

(3)



I am using the adapted version of the bracketed metrical grid introduced in chapter 2 to code prosodic structure. Each grid line is structured into feet (noted by the 'pipe'-symbol). A foot boundary at a given line in the grid implies a foot boundary on all lower lines. This leads to a strictly layered hierarchy of prosodic structure. The boundary rank, i.e., the strength of a given boundary, corresponds to the number of the grid line.

The data set used in this paper to test theories of prosodic phrasing consists of coordinate structures with 4 elements. There are exactly 11 different bracketings for such cases:

(4) a. Four elements at the top level:

Morgan and Joey and Norman and Ronny

b. Three elements at the top level:
Morgan or Joey or (Norman and Ronny)
Morgan or (Joey and Norman) or Ronny
(Morgan and Joey) or Norman or Ronny

c. Two elements at the top level:
Morgan or (Joey and Norman and Ronny)
(Morgan and Joey and Norman) or Ronny (Morgan and Joey) or (Norman and Ronny)

d. Two levels of embedding:
Morgan or (Joey and (Norman or Ronny))
Morgan or ((Joey or Norman) and Ronny)
(Morgan and (Joey or Norman)) or Ronny
((Morgan or Joey) and Norman) or Ronny

This data set serves to test theories of boundary assignment in prosody.

5.2.2 Two Theories

The LRB (Watson and Gibson (2004a,b)) relates the likelihood of an intonational phrase break at a given boundary to the complexity (measured in number of phonological phrases) of the two adjacent constituents:[1]

(5) The Left-Hand/Right-Hand Boundary (**LRB**) Hypothesis: The likelihood of a boundary between the LHS and RHS constituents increases with the size [counted in # of phonological phrases, MW] of each constituent. [due to processing cost of upcoming/preceding constituent; MW]

For the examples above, a stronger boundary occurs exactly where a complex constituent (underlined in the following example) follows (1a) or precedes (1b). The stronger boundaries are those represented by pipes on the top grid line.

[1]Effects of constituent length of the likelihood of a prosodic break based on impressionistic data have been observed as early as Bierwisch (1966a) and Martin (1971).

(6) Following and Preceding Complex Constituents

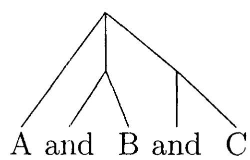
a. A or (B and C) b. (A or B) and C

```
| ×  | ×     ×  |     | ×     ×  | ×  |
| ×  | ×  |  ×  |     | ×  |  ×  | ×  |
  A    B     C          A     B    C
```

One advantage of the LRB over previous theories, according to Watson and Gibson (2004a), is that it is compatible with incremental processing: it is sufficient to look at the two adjacent syntactic constituents in order to determine whether or not a certain type of boundary occurs.

In order to account for coordinate structures in which more than 2 conjuncts are connected with a 'flat' prosody, the LRB would have to assume a syntactic representation with n-ary branching. The prosodic breaks between the conjuncts are then predicted to be equal to one another.[2]

(7) Flat Tree, Flat Prosody




```
| ×     ×     ×  |
| ×  |  ×  |  ×  |
  A     B     C
```

The connector 'and' would be ignored in evaluating the complexity of constituents, since they do not form separate phonological phrases. At each boundary, both the preceding and following constituent are simplex. The likelihood increases with every additional phonological phrase within an adjacent constituent. The LRB assumes then that there is a notion of phonological phrasing that itself is explained by some other factor, and makes predictions about the likelihood of intonational boundaries.

The complete predictions of the LRB for the data set discussed here (the data set in (4)) is summarized in table 5.1.

The alternative theory to be considered here is one in which the rank of a boundary reflects the syntactic embedding:

[2]The connectors themselves form a constituent with the following conjunct. See chapter 2 for discussion of the syntactic representation of coordinate structures, and for a discussion of unbounded vs. binary branching.

		A		B		C		D
1	A or B or C or D		1		1		1	
2	A or (B and C) or D		2		1		2	
3	A or B or (C and D)		1		2		1	
4	(A and B) or C or D		1		2		1	
5	(A and B) or (C or D)		1		3		1	
6	(A and B and C) or D		1		1		3	
7	A or (B and C and D)		3		1		1	
8	((A or B) and C) or D)		1		2		3	
9	(A and (B or C)) or D		2		1		3	
10	A or ((B or C) and D)		3		1		2	
11	A or (B and (C or D))		3		2		1	

Table 5.1: *Predictions of the LRB*

(8) Scopally Determined Boundary Rank (**SBR**):

If Boundary Rank at a given level of embedding is n, the rank of the boundaries between constituents of the next higher level is n+1.

In coordination structures, conjuncts/disjuncts at the same level are separated by boundaries of the same rank; if further embedding is introduced, boundaries of lower ranks encode the boundaries between the further nested constituents.

Again, if we use unbounded branching trees, there can be more than 2 constituents at one level of embedding, so coordinate structures with flat prosodies receive a prosody with equal boundaries separating the conjuncts.

In chapter 2, the boundary ranks of the SBR are derived by a cyclic syntax, using only binary branching trees. A 'flat' coordinate structure in which each conjunct is set off by a boundary of equal rank is one that is created in one single cycle; an 'articulated' coordinate structure with several different prosodic boundary ranks is one that was created in separate cycles.

The choice between binary branching trees with a derivational syntax and unbounded trees is not relevant for this chapter, and I will not consider it in the following.

The predictions for relative boundary strength for the examples in (6) and (7) are the same for the LRB and the SBR.

The SBR, however, predicts long distance effects for more complicated cases. The complexity of a constituent non-adjacent to a boundary can influence its relative rank.

Consider the following two examples, in particular the rank of boundary at the point where I put the placeholder '◇':

(9) a. A ◇ or B || or (C | and D)

b. (A | and B) || or C ◇ or D

The LRB predicts that the boundaries at '◇' are weak, since both adjacent constituents are simplex. The SBR predicts that these boundaries are strong, since they separate constituents at the higher level of embedding. This global long-distance effect would seem to be incompatible with incremental structure building.

The complete prediction for the data set in (4) are summarized in table 5.2. The cells in which the LRB and SBR differ are highlighted by boldface:

		A B C D		
1	A or B or C or D	1	1	1
2	A or (B and C) or D	2	1	2
3	A or B or (C and D)	**2**	2	1
4	(A and B) or C or D	1	2	**2**
5	(A and B) or (C or D)	1	**2**	1
6	(A and B and C) or D	1	1	**2**
7	A or (B and C and D)	**2**	1	1
8	((A or B) and C) or D)	1	2	3
9	(A and (B or C)) or D	2	1	3
10	A or ((B or C) and D)	3	1	2
11	A or (B and (C or D))	3	2	1

Table 5.2: *Predictions of the SBR*

This paper presents evidence that (i) long-distance effects are real, as predicted by the SBR; (ii) the SBR is compatible in principle with incremental structure building; and (iii), boundaries are scaled in strength relative to already produced boundaries.

In a production experiment, the following types of dialogues were presented using powerpoint:

(10) Experimenter: Who will solve the problem first?

Subject: Morgan and Joey are working together. Norman is working on his own. and so is Ronny. So one of the following will solve it first:

Morgan and Joey or Norman or Ronny.

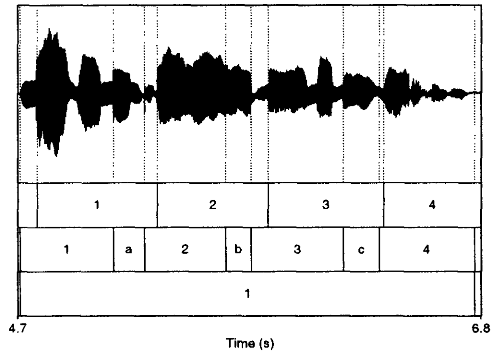


Figure 5-1: *Annotation Scheme*

The task was made easier by visual illustrations of the situations, further clarifying the grouping. The entire dialogue was recorded. The fragment answer (in boldface) was repeated three times; two sets of proper names were used for each of the 11 bracketings. 4 unpaid subjects were recorded. The data were annotated for word boundaries (second tier of annotation in (Fig. 5-1), and durations were measured using the Praat speech software (Boersma and Weenink, 1996).

The length of each constituent was normalized relative to the total duration of the answer. The mean of the three repetitions was computed, and constituent length was taken as a measure of the strength of the upcoming boundary, under the uncontroversial assumption that final pre-boundary lengthening increases with boundary strength (Price et al. (1991), Wightman et al. (1992) and references therein).

5.2.3 Syntactic Long-Distance Effects

Are there global effects of syntax on prosody that do not reduce to effects of adjacent constituents? In the following I will present evidence from a production experiment that there are. The evidence is in tune with the SBR but presents a problem for the LRB.

Effects of Adjacent Constituents

The LRB was designed to predict the likelihood of an intonational/intermediate phrase break[3]. The present study looked at boundary strength as reflected by duration of the preceding constituent. The advantage of this measure is that the annotation does not presuppose a theory of phrasing, and no labeling of prosodic categories (such as intonational phrase or intermediate phrase as in a ToBI-labeling) is necessary.

Watson and Gibson (2004a,b) used a different measure, the likelihood of intonational phrase break at a given boundary. This section presents evidence that the measure used here taps into the same phenomenon. Quantitative durational effects reflect the length of adjacent constituents. The results confirm an effect of adjacent constituents reported in Watson and Gibson (2004a,b). While Watson and Gibson (2004a,b) used the likelihood of intonational boundaries as the dependent measure, the quantitative measure of duration effect investigated.

First the effect of the length of the constituent following the boundary was tested by looking at the boundary strength after the first proper name in the structures in table 5.2.3.

			LRB	SBR	Cond
Baseline	1	A ♢ and **B** and C and D	1	1	0
	2	A ♢ or **B** or C or D	1	1	0
Simplex	3	(A ♢ and **B**) or C or D	1	1	-1
	4	(A ♢ and **B** and C) or D	1	1	-1
Complex	5	A ♢ or (**B and C**) or D	2	2	1
	6	A ♢ or (**B and C and D**)	3	2	1

Table 5.3: *Simplex vs. Complex Constituent Following the Boundary*

In the structures (1–4) in table 5.2.3, the constituent following the first connector is simplex and consists of only one proper name. In (5–6), the constituent following the first connector is complex and consists of 2 and 3 proper names respectively. The length of the first constituent (A), normalized relative to the overall duration of the utterance, was used as a measure of the strength of the following boundary.

[3]The two levels were conflated since inter-annotator reliability did not warrant making the distinction.

The predictions of the two theories for the three conditions 'Baseline', 'Simplex', and 'Complex' are the same: both SBR and LRB predict 'Baseline' and 'Simplex' to the same, and 'Complex' to be different from each of them.[4]

The omnibus ANOVA for the normalized length of constituent depending on the variable 'complexity' was highly significant F (1,4) = 85.13, $p < 0.000$. The means are plotted with the standard error in Fig. 5-2. The labels -1, 0, 1 reflect the value in the column 'COND' in table 5.2.3.

To test the effect of constituent complexity, pairwise post-hoc comparisons (Tukey) were conducted. Each pairwise comparison between 'baseline', 'simplex', and 'complex' was significant.

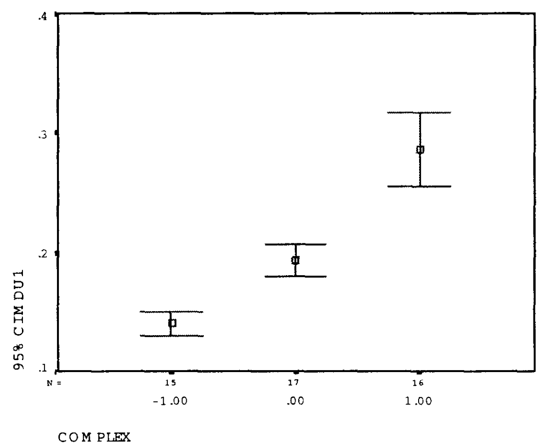


Figure 5-2: *Complexity of Following Constituent*

The contrasts between the conditions 'baseline' and 'simplex' on the one hand and 'complex' on the other are as expected by the LRB, and replicates the results of Watson and Gibson (2004a). They are also expected under the SBR.

However, the observed difference between 'baseline' and 'simplex' is unexpected both for the LRB and the SBR. In both conditions, the two constituents adjacent to the first boundary are simplex, and a boundary of equal strength is expected for the

[4]There is a difference for the predictions of the two theories with respect to the items within condition 'complex' (5 *vs.* 6): the LRB predicts an effect of the length of the following constituent, while the SBR does not. I will return to this difference between the SBR and the LRB in section 5.3.

LRB; they are on the lowest level of embedding, so a boundary rank of 1 is expected for the SBR.

The results for the two conditions differ, however. It matters whether or not *later in the utterance* there is a stronger boundary coming up (as in 3,4) or not (as in 1,2). This constitutes a long-distance effect, and shows that it is *not* sufficient to only consider two adjacent constituents at a given boundary. I will return to this effect in 5.3.

A second test was conducted to evaluate whether the complexity of the constituent *preceding* the boundary matters. Here, we are consdering the boundary that separates the last conjuncts. The boundary strength was again measured by relative pre-boundary lengthening, that is in this case lengthening of the proper name 'Norman', which directly precedes the boundary. The data is summarized in table 5.4.

			LRB	SBR	Cond
Baseline	1	A and **B** and C ◇ and D	1	1	0
	2	A or **B** or C ◇ or D	1	1	0
Simplex	3	A or B or (**C** ◇ and D)	1	1	-1
	4	A or (B and **C** ◇ and D)	1	1	-1
Complex	5	A or (**B and C**) ◇ or D	2	2	1
	6	(**A and B and C**) ◇ or D	3	2	1

Table 5.4: *Simplex vs. Complex Constituent Preceding the Boundary*

In the structures (1–4) in table 5.4, the constituent preceding the last connector is simplex and consists of only one proper name. In (5–6), this constituent is complex and consists of 2 and 3 proper names respectively. This time, the normalized length of the first constituent (C) was used as a measure of the strength of the following boundary.

The omnibus ANOVA for the normalized length of constituent depending on condition was highly significant $F(1,4) = 19.67$, $p < 0.002$. The means are plotted with the standard error in Fig. 5-3. The labels -1, 0, 1 reflect the value in the column 'COND' in table 5.4.

Pairwise post-hoc comparisons were conducted to test whether the three conditions were statistically different. As predicted by the LRB, both 'Baseline' and

'Simplex' were different from 'Complex'. The SBR also predicts this difference.

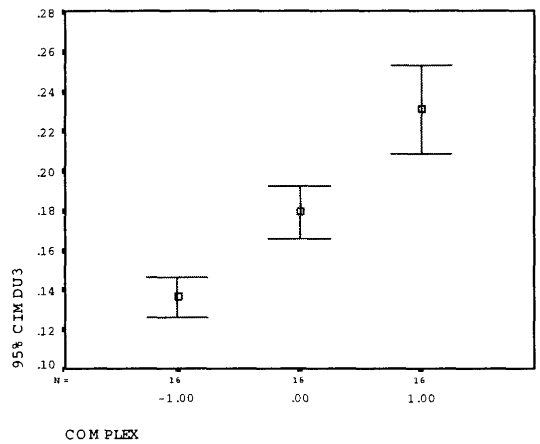


Figure 5-3: *Complexity of Preceding Constituent*

In addition, there was a significant difference between 'Baseline' and 'Simplex'. It matters whether or not a stronger boundary precedes in the structures. Once again, this effect is unexpected by the either the LRB or the SBR. And once again, it constitutes a long-distance effect.

The results presented so far confirm the claim in Watson and Gibson (2004a) that boundary strength is influenced by the complexity of adjacent constituents. While Watson and Gibson (2004a) were concerned with the likelihood of an intonational phrase break, the measure used in this study is quantitative lengthening.

The tests also showed initial evidence for long-distance effects. The effects observed here were not predicted by either the LRB or SBR. I will return to the question of how they can be explained, and suggest that the SBR can account for them once we understand boundary rank as a relative rather than an absolute notion.

5.2.4 Effects of Non-Adjacent Constituents

One major difference between the SBR and the LRB resides in the predictions for the examples in (9), repeated below:

(9) a. A ♢ or B || or (C | and D)

b. (A | and B) or || C ♢ or D

The LRB predicts a weak boundary at '◇' in (9):

(11) a. Morgan | or Joey || or Norman | and Ronny.

b. Morgan | and Joey || or Norman | or Ronny.

The SBR predicts a strong boundary:

(12) a. Morgan || or Joey || or Norman | and Ronny.

b. Morgan | and Joey || or Norman || or Ronny.

The data set used to decide between theories and the coding of conditions is summarized in Table (5.5).

													SBR	LRB	Cond
1		A	◇	or		B		or		C		◇ or	D	1	1	11
2		A		or		B		or		C	◇	or	D	1	1	11
3		A	◇	or		B		or	(	C		and	D)	2	1	21
4	(	A		and		B	)	or		C	◇	or	D	2	1	21
5		A	◇	or	(	B		and		C)		or	D	2	2	22
6		A		or	(	B		and		C)	◇	or	D	2	2	22

Table 5.5: *Data set for Testing Long-Distance effect*

The omnibus ANOVA for 'CND' was highly significant: $F(2,6) = 57.27$; $p < 0.000$. The means are plotted against the standard error in Fig. (5-4). The label '11' means that in this condition both theories predict a boundary of rank 1 and label '22' means that both theories predict a label of rank 2. The interesting condition is '21'. Here, the SBR predicts a boundary of rank 2, and the LRB a boundary of rank 1.

Post-hoc Tukey comparisons show differences between the 11 vs. 21 ($p <0.045$) and 11 vs. 22 ($p < 0.008$), but not between 21 vs. 22 ($p < .66$) (cf. Fig. (5-4)). The results are as predicted by the SBR. The boundaries at '◇' in (9) are strong, and they are strong due to the complexity of a non-adjacent constituent.

There are syntactic long-distance effects on prosody. The LRB does not account for the effect of the complexity of a non-adjacent constituent, but the SBR does, although not in all cases.

The next section will show evidence that boundaries are scaled in strength relative to upcoming and preceding boundaries. These results will then be used in section 5.3

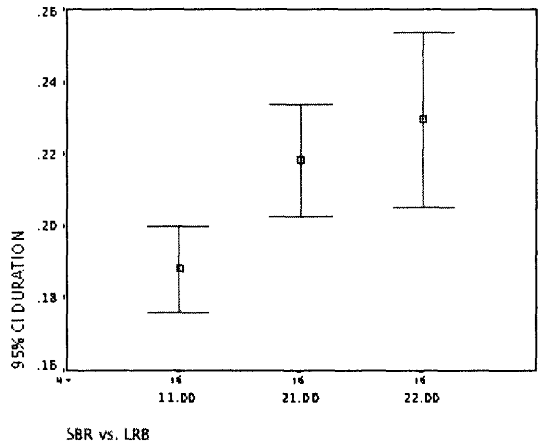


Figure 5-4: *Long Distance Effects*

to define a normalization based on the SBR, which is able to account for all of the effects reported in this section.

5.2.5 Looking Ahead

There is another kind of long-distance effect. In the realization of the first constituent, the complexity of the upcoming structure far beyond the adjacent constituent seems to play a role.

The metrical representation assigned in chapter 2 provides only relative boundary ranks. The phonetic and phonological implementation should reflect these boundaries, or at least should not contradict them. Durational cues are an important factor in the perceived boundary rank. If speakers anticipate the upcoming structure, then they might adjust the realization of the first constituent such that it allows them to realize the rest of the structure in such a way that the relative boundary ranks are correctly implemented within the phonetic range of their articulators.

The expectation is that boundaries are realized as stronger or weaker depending on how many lower or higher ranks will have to be realized later. The experiment was run in a way that certainly allowed the subjects to anticipate and plan the answer carefully. If there are look-ahead effects, they should be measurable in the length of the *first* constituent. Varying the length of the first constituent can be used to set up for upcoming stronger or weaker boundaries. The test-data used to probe for this effect are the following:

			BND
1	A or B or C or D	Baseline	0
2	A and B and C and D	Baseline	0
3	A or B or (C and D)	Lower Rank	-1
4	A or (B or C and D)	Lower Rank	-1
5	A or (B and (C or D)	Lower Rank	-2
6	(A and B) or C or D	Higher Rank	1
7	(A and B and C) or D	Higher Rank	1
8	(A or B) and C) or D	Higher Rank	2

Table 5.6: *Look-Ahead*

The omnibus ANOVA for BND was highly significant: $F(4,12) = 2.28$; $p < 0.000$. The plot of the means reveals that the length of the first constituent varies by and large as expected with the number of upcoming weaker/stronger constituents:

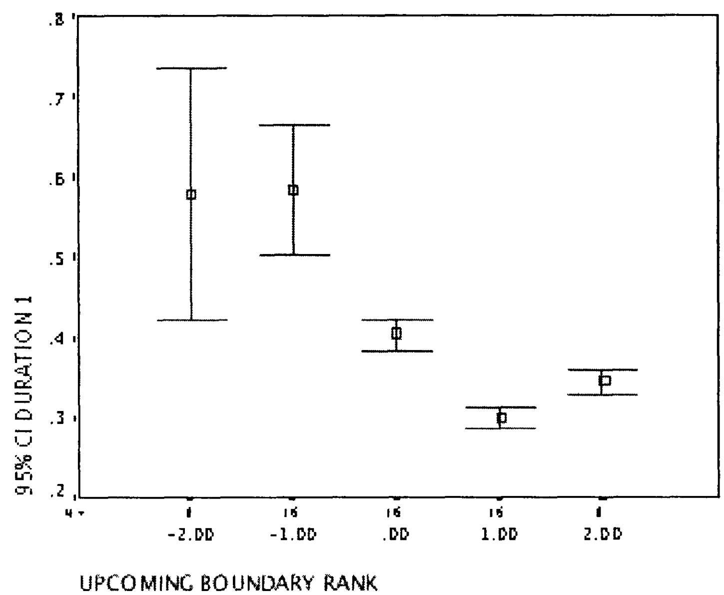


Figure 5-5: *Look-Ahead Effect*

This look-head effect might reflect speech planning, and thus be due to considerations that lie at the heart of the proposal in Watson and Gibson (2004a).

The LRB, however, predicts an effect that relates to the length of an upcoming constituent—independent of the degree of embedding. We can use the following data set to test whether this effect is real.

		LRB	SBR	Length
1	A or B or (C and D)	1	2	1
2	A or (B and C) and D	2	2	2
3	A and (B or C or D)	3	2	3

Table 5.7: *Length of the Upcoming Constituent*

The SBR predicts an equal boundary rank at the relevant boundary. The LRB predicts that the boundary should get stronger with an increasing length of the following constituent.

The omnibus ANOVA for the variable 'Length' approaches significance. $F(1,4) = 4.80$; $p < 0.056$. The means are plotted in figure 5-6. The means of the conditions summarized in the column NST in Table 5.7 are summarized in Fig. 5-6.

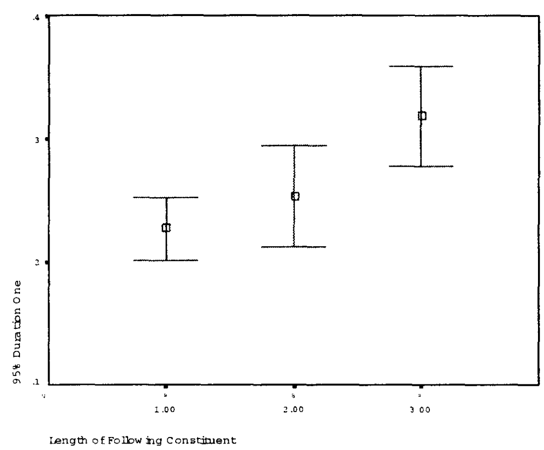


Figure 5-6: *Length of Following Constituent*

Just as predicted by the LRB, there does seem to be an effect of the length of the upcoming constituent, even if it is not quite significant. The effect is a look-ahead effect, in that it influences the realization of the first constituent, depending on the length of the second constituent.

5.2.6 Looking Back

The realization of prosodic structure may also be affected by the prosodic shape of structure produced earlier. Again, this effect takes more into account than just the preceding constituent.

Boundary strength scaling relative to an already produced boundary can be achieved by compressing or inflating constituents realized later in the sentence. The prediction is that the length of the *final* constituent should vary depending on whether or not it is in an early- or late-nesting structure. 'Early nesting' means that a constituent with a low level of syntactic embedding early in the utterance is followed by higher material; 'late-nesting' means that a constituent with a low level of embedding occurs late in the utterance. The data set to test for an effect of Boundary-Strength Scaling is summarized in table 5.8.

						NST
1	A or	B or	C or	D	Baseline	0
2	A and	B and	C and	D	Baseline	0
3	A or	B or	(C and	D)	Late Nesting	-1
4	A or	(B and	C and	D)	Late Nesting	-1
5	(A and	B) or	C or	D	Early Nesting	+1
6	(A and	B and	C) or	D	Early Nesting	+1

Table 5.8: *Late-Nesting vs. Early-Nesting and Look-Back*

The omnibus ANOVA for 'Nesting' approaches significance: $F(2,6) = 4.34$; $p < 0.068$, the means are as predicted (Fig. 5-7).

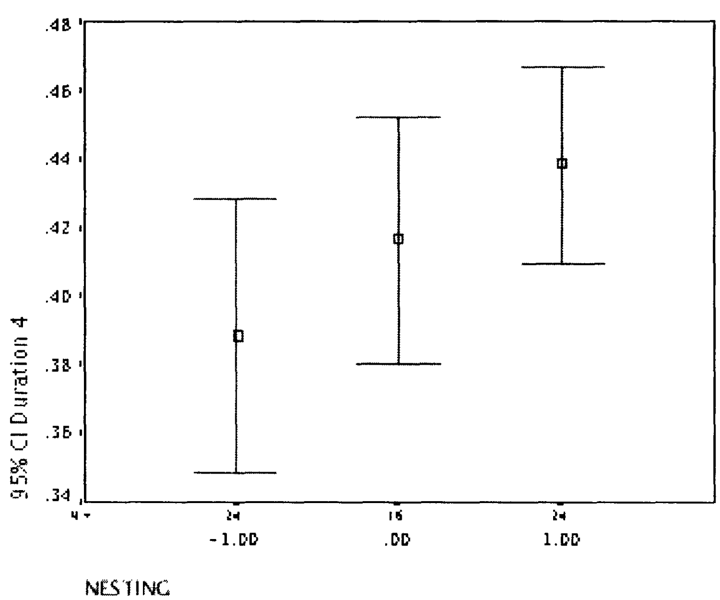


Figure 5-7: *Looking Back: Nesting*

The results are not quite significant. But if look-back effects are real, then boundaries can be scaled as stronger or weaker relative to already produced prosodic structure, in order to realize the relative boundary ranks that are imposed by the syntactic structure—this strategy can be used in the case of long-distance effects, if the look-ahead didn't make the preceding boundaries sufficiently strong in anticipation of the weaker boundary.

The variation in the duration of the *last* constituent is due to variation in the level of embedding of the last constituent—it is not due to pre-boundary lengthening. The last constituent is always at the end of the utterance, therefore the following boundary should always be a strong one.

There is a look-back effect that relates to the rank of the preceding boundary. The LRB predicts an effect that relates to the length of the preceding constituent, independent of the degree of embedding. We can use the data set in table 5.9 to test whether this effect is real.

												LRB	SBR	Length
1	(	A	or		B	)	or	C		◇	and D	1	2	1
2		A	or	(	B		and	C	)	◇	and D	2	2	2
3	(	A	and		B		or	C	)	◇	or D	3	2	3

Table 5.9: *Length of the Preceding Constituent*

The SBR predicts equal boundary ranks at the relevant boundary ('◇') in the three structures. The LRB predicts that the boundary should get stronger with an increasing length of the preceding constituent.

The length effect is significant. The omnibus ANOVA for the variable 'Length' shows a clear effect. $F(1,4) = 13.12$; $p < 0.006$. The means are plotted in figure 5-8.

As predicted by the LRB, there is an effect of the length of the upcoming constituent. But note that the means do not relate the length of the relevant constituent with the length of the preceding constituents in a straightforward way. This suggests we are not not looking at the data in the right way yet. The remainder of the paper presents a different way of looking at the data, which captures all of the effects reported so far.

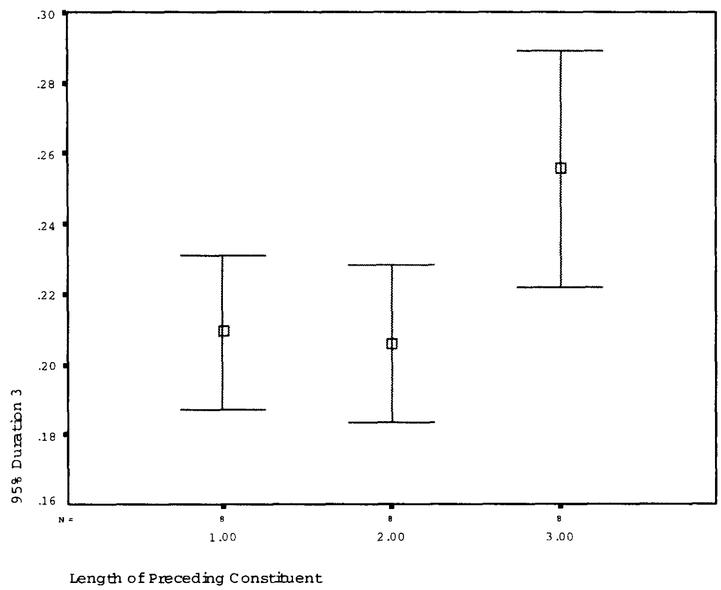


Figure 5-8: *Looking Back: Length*

5.3 Left-to-Right Scaling

What is a good model of the various kinds of lengthening effects observed in this chapter so far? The SBR predicts some but not all long-distance effects, the LRB predicts length-based effects, but fails to predict any long-distance effects. Both theories are not able to explain the full range of data.

Watson and Gibson (2004a) argue that since the LRB uses only local information, it is compatible with incremental structure building, and is thus superior to global syntactic approaches such as Cooper and Paccia-Cooper (1980), Gee and Grosjean (1983), Ferreira (1988). One premise of this argument is the assumption of a fixed prosodic hierarchy:

(13) Prosodic hierarchy (Selkirk, 1986, 384):

```
(____________________________________________________________)  Utt
(____________________________________)(_____________________)  IPh
(__________)(________________________)(_____________________)  PPh
(__________)(_______)(_______________)(____________)(_______)  PWd
(__)(______)(______)(__)(___________)(________)(__)(______)  Ft
(__)(__)(__)(__)(__)(__)(__)(__)(__)(__)(__)(__)(__)(__)(__)  Syl
```

Under this assumption, a boundary is 'stronger' if it has the label of a prosodic constituent higher on the hierarchy, and 'weaker' with a lower label. It is crucial then to start out with the right level of prosodic constituent, so that one does not run out of categories in order to produce stronger or weaker boundaries later in the sentence. To illustrate the point, consider early- vs. late-nesting structures:

(14) a. Early Nesting:
(Morgan and Joey) or Norman or Ronny

b. Late Nesting:
Morgan or Joey or (Norman and Ronny)

In (14a), starting out with an intonational phrase leading to an intonational phrase break after 'Morgan' would make it impossible to signal a stronger boundary after 'Joey'—unless one placed an utterance end at this point. Conversely, in (14b), starting out with a phonological phrase break after 'Morgan' would make it impossible to signal a weaker boundary after 'Norman'—unless one mapped 'Norman and Ronny' to a single prosodic word.

The alternative to the prosodic hierarchy is a recursive Prosodic System, in which boundaries are ranked according to a metrical representation, without a difference in categorical label between the different ranks, as proposed in chapter 2. Tones are mapped to grids indirectly, obeying alignment principles, in which the top gridline plays a crucial role (Liberman (1975), Gussenhoven (1990)). The assumption here is that exactly all top-line grid-marks are realized as pitch accents, or at least other accents are severely reduced in pitch range.

(15)
$$\begin{array}{ccccccccc}
| & \times & & \times & & \times & & \times & | \\
| & \times & | & \times & & \times & & \times & | \\
| & \times & | & \times & | & \times & & \times & | \\
| & \times & | & \times & | & \times & | & \times & | \\
 & \mathrm{A} & & \mathrm{B} & & \mathrm{C} & & \mathrm{D} &
\end{array}$$

This system allows for ***Boundary-Strength Scaling*** (BSS), in other words, boundaries can be scaled as stronger or weaker relative to other boundaries. Carlson et al. (2001) indeed report an effect that shows that two boundaries are adjusted relative to each other in the disambiguation of attachment ambiguities.

What is crucial in the theory proposed in chapter 2 is the relative rank of a boundary, not its categorical identity. BSS could be achieved either by look-ahead, which would assure that early boundaries are weaker/stronger than upcoming boundaries, or by look-back, which would scale a boundary as weaker/stronger relative to an *already produced* boundary.

This section shows that by normalizing the length of each constituent relative to the first constituent, and by coding boundary rank relative to the boundary following the first constituent (using the SBR), we can substantially improve the coverage of the data.

The idea is to normalize the data in such a way as would make sense if speech production is indeed incremental and left-to-right. The idea of a left-to-right organization of speech has been supported in Phillips (1996, 2003) who proposes that grammatical derivations as well should incrementally build structure in in a left-to-right fashion.

The left-to-right implementation of the SBR proposed here accounts for the long-distance effects, for 'looking-back'-effects, and also for the effects of constituent length. Furthermore, it allows us to define a metric of look-ahead effects.

5.3.1 A Normalization

The look-ahead effects, as reflected in the length of the first constituents, reflect a speaker's anticipation of the complexity of upcoming structure.

The observations in the previous section suggest another factor is at play. It looks as if boundaries are scaled relative to upcoming or preceding boundaries, in order to assure that they have the right relative rank. If the system really operates with relative boundary rank rather than absolute boundary strength, it would make sense for a speaker to start out strong when weaker boundaries are coming up and vice versa. The subsequent boundaries are then scaled relative to the first boundary accordingly.

The purpose of the look-ahead variation in the first constituent is to assure that the relative ranks can be realized given the physical range that a speaker has available to articulate relative boundary strength; it may also have the purpose of assuring that the produced boundary ranks are perceptually distinct.

If this is how the system works, we should first look at the data by controlling for the variation in the first constituent, and then verify whether the rest of the structure is indeed scaled relative to that first constituent.

A simple way of doing so is by normalizing the duration of all constituents relative to the first one. This can be done by dividing the length of every constituent by that of the first constituent. The length of the first constituent is then always 1—in other words, we have neutralized the look-ahead effect.

Now we should be able to observe a relative scaling of all other material relative to the first constituent. A constituent is going to be longer or shorter depending on whether the boundary rank that follows it is stronger or weaker than the boundary rank following the first constituent. This presupposes a theory of boundary rank assignment. I will use the SBR here, and show that it allows for a good model of the observed data.

The boundary ranks predicted by the SBR are recoded in such a way that it reflects boundary relative to the first boundary, which is the baseline or the 'zero boundary rank'.

The predictions of the left-to-right version of the SBR, or normalized SBR (henceforth NSBR), are summarized in table 5.10. The table gives the same ranks as the SBR, but instead of giving an absolute value, each rank is computed *relative to the rank following the first constituent.*

		A	B	C	D
1	A or B or C or D		0	0	0
2	A or (B and C) or D		0	-1	0
3	A or B or (C and D)		0	0	-1
4	(A and B) or C or D		0	1	1
5	(A and B) or (C or D)		0	1	0
6	(A and B and C) or D		0	0	1
7	A or (B and C and D)		0	-1	-1
8	((A or B) and C) or D)		0	1	2
9	(A and (B or C)) or D		0	-1	1
10	A or ((B or C) and D)		0	-2	-1
11	A or (B and (C or D))		0	-1	-2

Table 5.10: *NSBR*

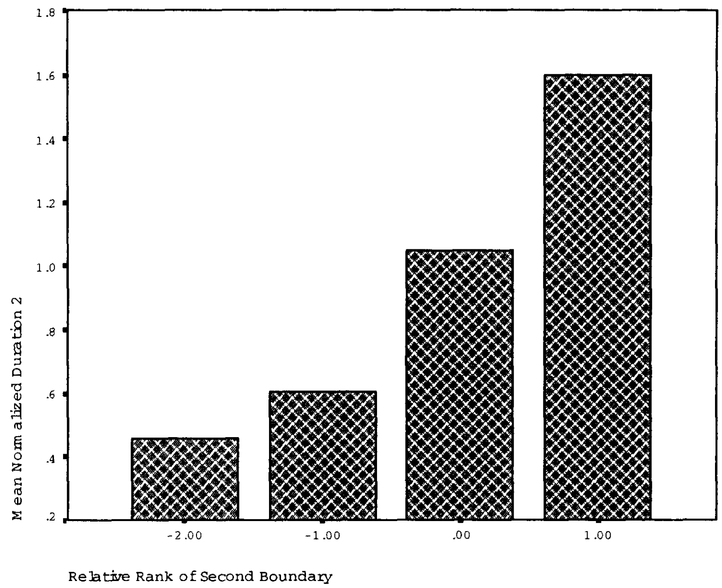


Figure 5-9: *Normalized Second Constituent*

In the following, I will illustrate some of the predictions of the new normalization, and illustrate that it provides a good model of how speakers scale boundaries.

Consider the graph in figure 5-9, which plots the means of the second constituent by the relative boundary rank.

Post-hoc comparisons show significant differences between three different levels, as shown in figure 5-10.

The left-to-right normalization proposed here is a substantial improvement over the original SBR, where normalization was achieved by dividing by the duration of the entire utterance. This becomes apparent when we compare the two theories by looking at the mean durations of the relevant constituent.

The original SBR predicts only two different ranks for the second boundary. The left-to-right SBR, on the other hand, predicts five different categories, and allows for a finer-grained understanding of the data.

A look at the third conjunct is similarly encouraging. Consider the means of the normalized length of the third conjunct, plotted against the strength of the following boundary in figure 5-11.

Post-hoc comparisons distinguish four different categories of boundary ranks (fig-

Normalized Duration of Second Constituent

	DBND1	N	Subset 1	Subset 2	Subset 3
Tukey HSD[a,b,c]	-2.00	4	.4609		
	-1.00	28	.6060		
	.00	32		1.0500	
	1.00	24			1.6012
	Sig.		.713	1.000	1.000
Tukey B[a,b,c]	-2.00	4	.4609		
	-1.00	28	.6060		
	.00	32		1.0500	
	1.00	24			1.6012

Means for groups in homogeneous subsets are displayed.
Based on Type III Sum of Squares
The error term is Mean Square(Error) = .104.

a. Uses Harmonic Mean Sample Size = 11.154.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

Figure 5-10: *3 Subsets defined by Post-hoc Comparisons*

ure 5-12).

The left-to-right normalization makes accurate predictions for the length of the conjuncts. In the following, I will illustrate how it can explain the long-distance effects and the various look-ahead and looking-back effects discussed above. The NSBR is a promising model of durational effects in prosodic structure. It is crucially based on the theory of metrical structure presented in chapter 2, which assigns a relative metrical structure that provides the relative boundary ranks.

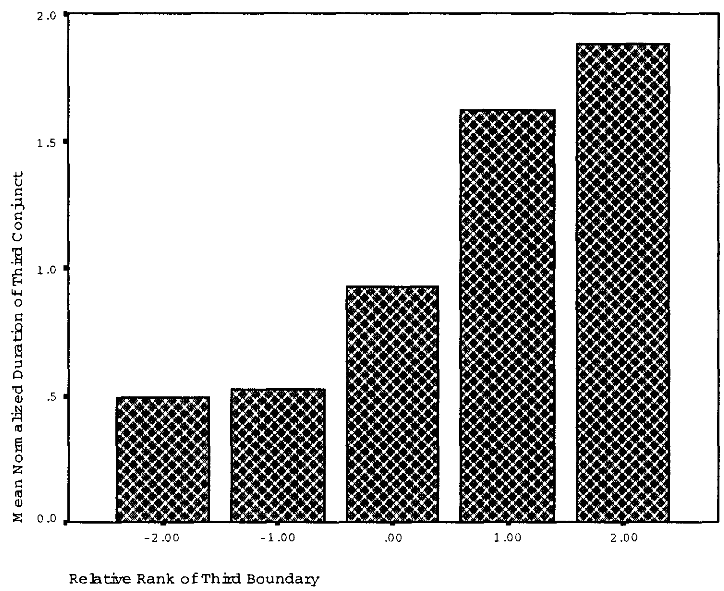


Figure 5-11: *Normalized Length of Third Constituent*

Normalized Duration of Third Constituent

	Rank Third	N	Subset 1	2	3	4
Tukey HSD[a,b,c]	-2.00	8	4955			
	-1.00	20	.5266			
	00	32		.9317		
	1 00	20			1 6244	
	2 00	8				1 8838
	Sig.		996	1 000	1 000	1 000
Tukey B[a,b,c]	-2.00	8	4955			
	-1.00	20	5266			
	.00	32		.9317		
	1 00	20			1.6244	
	2 00	8				1.8838

Means for groups in homogeneous subsets are displayed
Based on Type III Sum of Squares
The error term is Mean Square(Error) = 4 47 3E-02

a Uses Harmonic Mean Sample Size = 13 115

b The group sizes are unequal The harmonic mean of the group sizes is used. Type I error levels are not guaranteed

c Alpha = 05

Figure 5-12: *4 Subsets defined by Post-hoc Comparisons*

5.3.2 Looking Ahead Again

The NSBR normalizes length according to the duration of the first constituent, thus keeping the length of the first constituent at the value 1 across conditions. Of course, as we have already seen, there is variation in the length of the first constituent.

The NSBR is only complete if we can find an insightful way of explaining this variation as well. I propose that a speaker aims to adjust the duration of the first constituent such that that the upcoming boundaries can be scaled relative to the first boundary within the limited range of possibilities that are available due to articulatory and perceptual limits. So if a boundary has to be stronger than upcoming structure, it makes sense to start out strong, but if stronger boundaries are coming than it makes start out weak.

A successful look-ahead strategy would then be one that counts the number of upcoming stronger and weaker boundaries, and adjusts the strength of the first boundaries accordingly. The strength of the first boundary is reflected in the length of the first constituent.

In particular, one possibility is that the look-ahead measure used by a speaker who does 'perfect look-ahead' is exactly the sum of all upcoming boundary ranks.

This is only one of many possible measures of look-ahead though. While the boundary ranks we tested in the previous section were assigned by the theory developed in chapter 2, the theory gives us no expectation whatsoever with respect to what measures look-ahead best. The purpose of this section is then to show that there are ways to account for look-ahead within the NSBR—but the discussion remains speculative.

Table 5.11 summarizes the sums of upcoming boundaries after the first one for each of the 11 conditions. The inverse of this value is the look-ahead value (LAV) for a particular structure.

If the LAV is positive, then the first constituent will be longer, if it is negative, then it will be shorter compared to the cases where the LAV is zero.

The theory predicts 6 different LAVs. Plotting the mean duration of the first

		A	B	C	D	Sum	LAV
1	A or B or C or D	0	0	0		0	0
2	A or (B and C) or D	0	-1	0		-1	1
3	A or B or (C and D)	0	0	-1		-1	1
4	(A and B) or C or D	0	1	1		2	-2
5	(A and B) or (C or D)	0	1	0		1	-1
6	(A and B and C) or D	0	0	1		1	-1
7	A or (B and C and D)	0	-1	-1		-2	2
8	((A or B) and C) or D)	0	1	2		3	-3
9	(A and (B or C)) or D	0	-1	1		0	0
10	A or ((B or C) and D)	0	-2	-1		-3	3
11	A or (B and (C or D))	0	-1	-2		-3	3

Table 5.11: *Look-Ahead Values*

constituent against this variable (figure 5-13) illustrates that the correlation is fairly good—except for the extreme values, where the correlation breaks down.

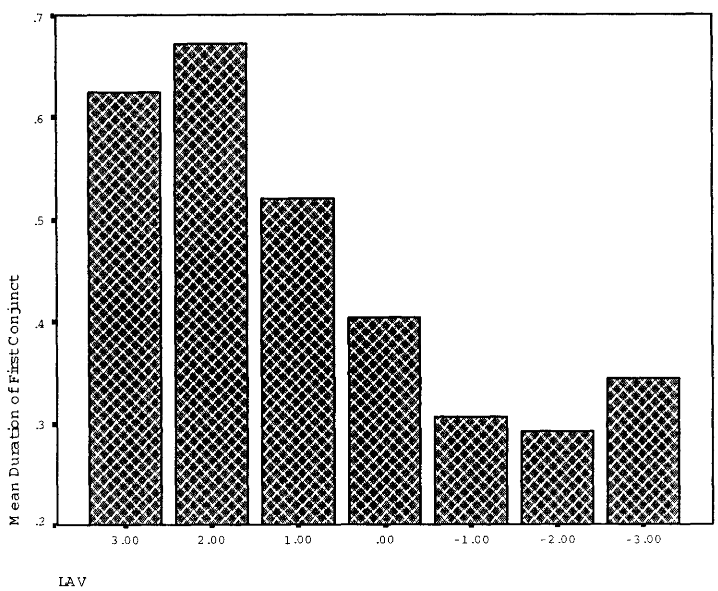


Figure 5-13: *LAV*

We can improve the correlation by the following stipulation: look-ahead only consider the structure of one cycle ahead. This is a reasonable stipulation if each cycle constitutes a processing unit, and look-ahead is constrained by how many processing units it can span. The predictions then change as outlined in table 5.12.

The means of the duration of the first constituent are again plotted against the expected look-ahead-value in figure 5-14.

									A	B	C	D	Sum	LAV
1	A or		B	or	C	or	D		0	0	0		0	0
2	A or	(	B	and	C)	or	D		0	-1	0		-1	1
3	A or		B	or (	C	and	D)		0	0	-1		-1	1
4	(A and		B)	or	C	or	D		0	1	1		2	-2
5	(A and		B)	or (	C	or	D)		0	1	0		1	-1
6	(A and		B	and	C)	or	D		0	0	1		1	-1
7	A or	(	B	and	C	and	D)		0	-1	-1		-2	2
8	((A or		B)	and	C)	or	D)		0	1	2		1	-1
9	(A and	(	B	or	C))	or	D		0	-1	1		1	-1
10	A or	((	B	or	C)	and	D)		0	-2	-1		-1	1
11	A or	(	B	and (	C	or	D))		0	-1	-2		-1	1

Table 5.12: *Sum of Boundaries Up and Including Next Cycle*

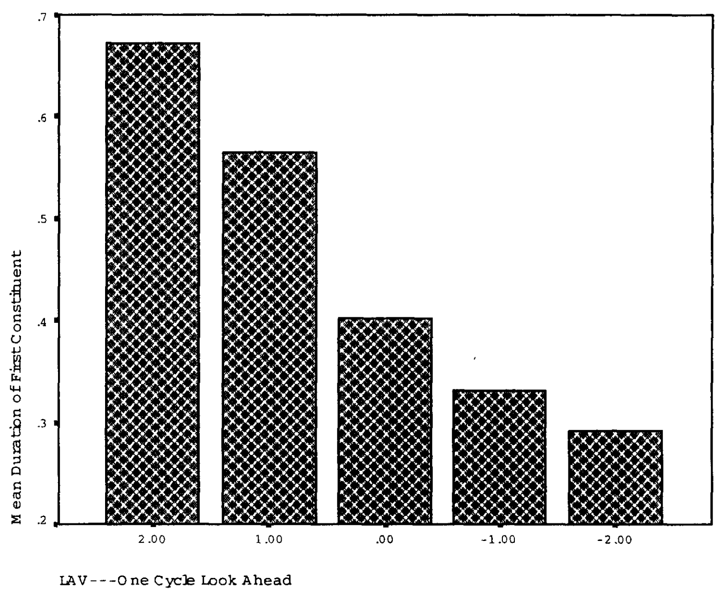


Figure 5-14: *LAV, 2 Cycle Look-Ahead*

The revised LAV theory then predicts 5 different boundary ranks. We can test this by running post-hoc comparisons on the mean duration of the first constituent in those 5 conditions. A Tukey-Post-B subsets test differentiates 4 categories in look-ahead effects, as shown in figure 5-15.

The NSBR also accounts for the apparent effects of the length of the following constituent. This effect was predicted by the LRB, but not by the original version of the SBR. The NSBR in conjunction with the look-ahead value makes the predictions summarized in table 5.3.2.

Mean Duration of First Conjunct

	LAVCYC	N	Subset 1	2	3	4
Tukey HSD[a,b,c]	-2.00	8	.2926			
	-1.00	28	.3323			
	.00	16	.4032			
	1.00	28		.5656		
	2.00	8		.6716		
	Sig.		.065	.084		
Tukey B[a,b,c]	-2.00	8	.2926			
	-1.00	28	.3323	.3323		
	.00	16		.4032		
	1.00	28			.5656	
	2.00	8				.6716

Means for groups in homogeneous subsets are displayed.
Based on Type III Sum of Squares
The error term is Mean Square(Error) = 1.094E-02.

a. Uses Harmonic Mean Sample Size = 13.023.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

Figure 5-15: *Post-Hoc Comparisons on LAV Effect*

										LRB	SBR	Length	LAV
1	A	or		B	or	(C	and	D)		1	2	1	1
2	A	or	(	B	and	C)	and	D		2	2	2	1
3	A	and	(	B	or	C	or	D)		3	2	3	2

Table 5.13: *Length of the Upcoming Constituent*

The omnibus ANOVA for the variable LAV is highly significant ($F(4,1) = 34.94$; $p < 0.011$.

Long-Distance Effects Revisited

The NSBR combined with look-ahead values (LAV) accounts for a long-distance effect that remained unexplained in both in the LRB and the SBR. Table 5.14 summarizes the predictions of the three theories.

										LRB	SBR	LAV	Cond
Baseline	1	A ◇	and	**B**	and	C	and	D		1	1	0	0
	2	A ◇	or	**B**	or	C	or	D		1	1	0	0
Simplex	3	(A ◇	and	**B**)	or	C	or	D		1	1	-2	-1
	4	(A ◇	and	**B**	and	C)	or	D		1	1	-1	-1
Complex	5	A ◇	or	(**B**	**and**	**C**)	or	D		2	2	1	1
	6	A ◇	or	(**B**	**and**	**C**	**and**	**D**)		3	2	2	1

Table 5.14: *Simplex vs. Complex Constituent Following the Boundary*

The look-ahead value defined by the NSBR serves to make distinction between the three conditions. We can add the LAV-values within each condition and end up with a three-way differentiation. This accounts for the data observed in figure 5-2.

In fact, however, the NSBR makes a 5-way distinction between the 6 structures. This is exactly reflected in the distribution of the means (figure 5-16).

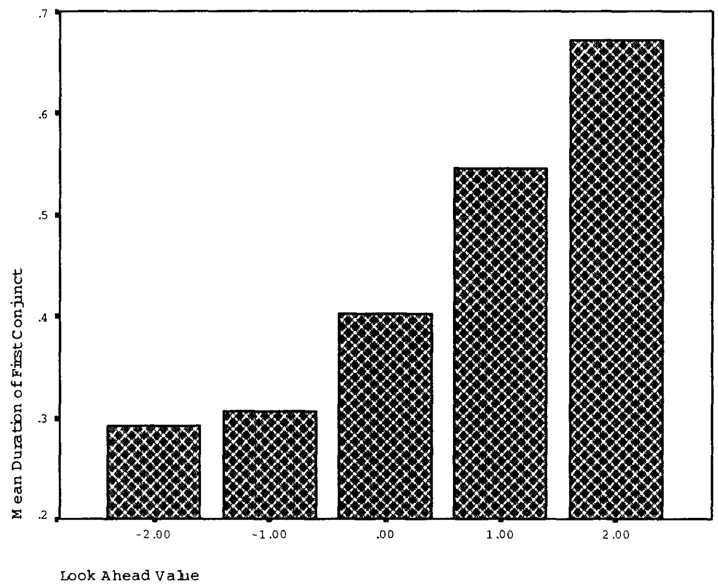


Figure 5-16: *5-Way Distinction Predicted by the LAV*

The predictions of the LAV are thus quite accurate even in a case where both LRB and SBR failed to account for the data.

The variation in length in the first conjunct is a look-ahead effect. The look ahead value (LAV) defined based on the sum of the upcoming boundary ranks, using the ranks defined by the NSBR, provides an adequate measure of the look-ahead actually observed. The theory accounts for a number of effects that go unaccounted both in the LRB- and the SBR-theory.

Consider now the predictions for the example that was problematic for the LRB, repeated below:

(9) a. A $\diamondsuit$ or B $||$ or (C $|$ and D)

b. (A $|$ and B) $||$ or C $\diamondsuit$ or D

The crucial boundary in example (a) is the one following the first conjunct. Since every constituent is normalized with respect to the first conjunct, no variation in length of that constituent is predicted based on the ranks assigned by the NSBR. The rank at ◊ is always zero. But based on the discussion in the last section, a look-ahead effect is predicted, depending on the sum of the upcoming boundaries, the LAV-value of the structures. The predictions for the variation in length before ◊ are summarized in table 5.15.

		SBR	LRB	LAV	Cnd
1	A ◊ or B or C or D	1	1	0	11
2	A ◊ or B or (C and D)	2	1	1	21
3	A ◊ or (B and C) or D	2	2	1	22

Table 5.15: *Data set for Testing Long-Distance effect*

The predictions of the NSBR with look-ahead are the *same* as those of the SBR: the duration of the first constituent should be higher for the conditions 21 and 22 compared to condition 11. This is borne out by the data, as discussed in section 5.2.4.

The case of (9b) is more complicated. Due to the new normalization scheme, a direct comparison between the three theories is not possible. Consider the predictions of the three theories (LRB, SBR, NSBR) in table 5.16.

		SBR	LRB	NSBR	Cnd
1	A or B or C ◊ or D	1	1	0	11
2	(A and B) or C ◊ or D	2	1	1	21
3	A or (B and C) ◊ or D	2	2	0	22

Table 5.16: *Data set for Testing Long-Distance effect*

In section 5.2.4, we saw that the SBR fares better than the LRB in predicting this pattern. The NSBR makes very different predictions. Condition 11 and 22 are predicted to be identical—thus differing in precisely the categories where both the LRB and the SBR made the same predictions. Since the data are normalized in a different way, we cannot directly compare the three theories.

We can test whether the predictions of the NSBR are borne out given the new normalization. Figure 5-17 illustrates that the means look as predicted.

Figure 5-18 illustrates that post-hoc comparisons confirm the predictions of the NSBR.

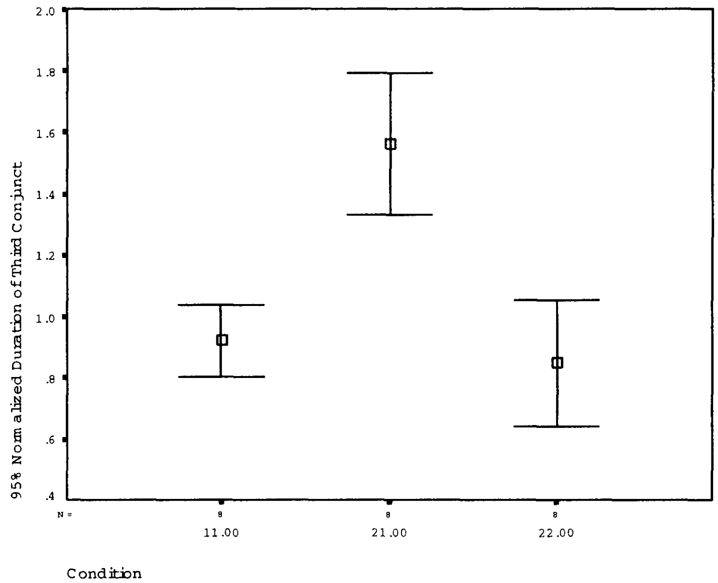


Figure 5-17: *Long Distance Effect in Normalized Data*

The results show that the NSBR makes correct predictions for the normalized length of the relevant third conjunct.

But what about the *actual* length of the third conjunct? The means of the third conjunct are summarized in figure 5-19—the third conjunct in conditions 2 and 3 have about equal length, while 1 is much shorter.

The reason the actual length does not reflect directly the relative rank of the NSBR is that the three structures in table 5.15 are associated with different look-ahead values. Only the look-ahead value of the third condition is positive, the others are zero. The different look-ahead value affect the length of the first conjunct, which in turn has an effect on the normalized length of the third conjunct. The actual length of the third condition is thus higher compared to the other two conditions than the graph in (5-17) suggests.

Normalized Duration of Third Constituent

	Condition	N	Subset 1	Subset 2
Tukey HSD[a,b]	22.00	8	.8491	
	11.00	8	.9223	
	21.00	8		1.5625
	Sig.		.655	1.000
Tukey B[a,b]	22.00	8	.8491	
	11.00	8	.9223	
	21.00	8		1.5625

Means for groups in homogeneous subsets are displayed.
Based on Type III Sum of Squares
The error term is Mean Square(Error) = 2.694E-02.

a. Uses Harmonic Mean Sample Size = 8.000.

b. Alpha = .05.

Figure 5-18: *Post-Hoc Comparisons on Look-Ahead Effect*

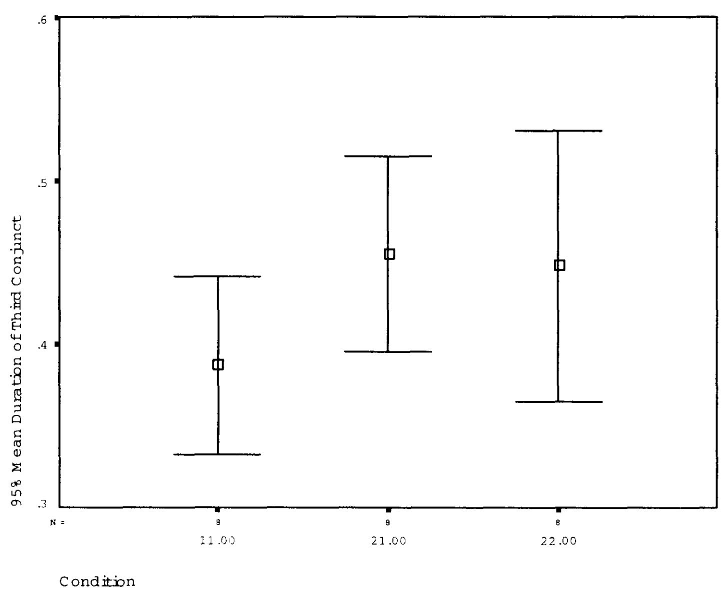


Figure 5-19: *Long Distance Effects*

5.4 The Prosody of Function Words

The comparisons so far only looked at length effects on the proper names in the structures. The reasoning was that the pre-boundary lengthening between conjuncts is a measure of the boundary strength. Another question of interest, however, is whether there are any length effects on the *coordinators.*

Coordinators are usually pro-cliticized[5], just like many other function words. Gee and Grosjean (1983, 433) simply assign a boundary of rank 0 to the boundary following them. This would predict however that there is no variation in the length of function words depending on where in the structure they occur. In the following example, e.g., 'or' and 'and' should have the same prosodic status:

(16) A or B or (C and D)

But within the relative theory of metrical structure in chapter 2, pro-cliticizing a function word may simply mean the following: the boundary separating it from the following conjunct is weaker than the boundary separating it from the preceding conjunct.

(17) Function Word Pro-Clititcization
If the boundary preceding a functor is of rank 1, then the boundary between the functor and its complement is n-1:

```
|          ×   |
|  ×   |   ×   |
   F     Host
```

In chapter 6, I will discuss how this 'prosodic subordination' of function words comes about. The prosodic structure assigned to (16) looks then as follows:

(18) Prosodic Structure with Functors:

```
|  ×  |        ×  |        ×           ×  |
|  ×  |  ×  |  ×  |  ×  |  ×           ×  |
|  ×  |  ×  |  ×  |  ×  |  ×  |        ×  |
|  ×  |  ×  |  ×  |  ×  |  ×  |  ×  |  ×  |
   A    or     B    or     C    and    D
```

[5] Although they may rebracket to the left, e.g. with 'n

The variance in the length of the connectors can then be conceived of as pre-boundary lengthening. The prediction is that the length of the connectors should co-vary with the length of the preceding constituent: If the preceding boundary is stronger, it means that the boundary separating the connector form the following conjunct is stronger. Figure 5-20 summarizes the means.

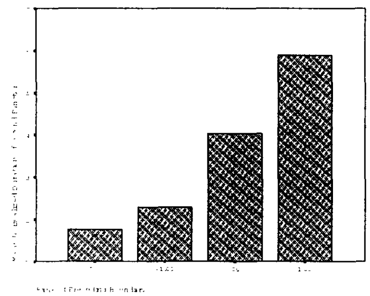


Figure 5-20: *Rank of Second Boundary, Second Functor*

Post-hoc Comparisons differentiate three different classes of functors, as illustrated in figure 5-21.

Mean Normalized Duration of Second Connector

	Bnd Rank	N	Subset 1	2	3
Tukey HSD[a,b,c]	-2.00	4	.1773		
	-1.00	28	.2302	.2302	
	.00	32		.4040	.4040
	1.00	24			.5890
	Sig.		.899	.112	.081
Tukey B[a,b,c]	-2.00	4	.1773		
	-1.00	28	.2302	.2302	
	.00	32		.4040	
	1.00	24			.5890

Means for groups in homogeneous subsets are displayed.
Based on Type III Sum of Squares
The error term is Mean Square(Error) = 3.235E-02.

a. Uses Harmonic Mean Sample Size = 11.154.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

Figure 5-21: *Statistically Different Boundary Ranks*

Similarly accurate predictions are made for the last connector. Figure 5-22 summarizes the means plotted against the rank of the preceding boundary.

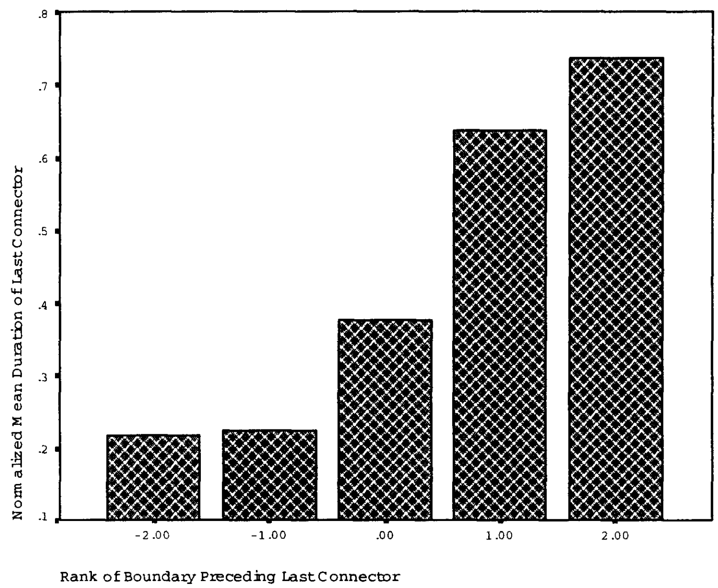


Figure 5-22: *Rank of Third Boundary, Last Functor*

Post-Hoc Comparisons differentiate 4 categories of lengthening for the last connector, as illustrated in figure 5-23.

A more extensive discussion of function words and how they receive their prosody within the theory outlined in chapter 2 will be discussed in chapter 6.

Normalized Mean Duration of Last Connector

	DBND2	N	Subset 1	2	3	4
Tukey HSD[a,b,c]	-2.00	8	.4955			
	-1.00	20	.5266			
	.00	32		.9317		
	1.00	20			1.6244	
	2.00	8				1.8838
	Sig.		.996	1.000	1.000	1.000
Tukey B[a,b,c]	-2.00	8	.4955			
	-1.00	20	.5266			
	.00	32		.9317		
	1.00	20			1.6244	
	2.00	8				1.8838

Means for groups in homogeneous subsets are displayed.
Based on Type III Sum of Squares
The error term is Mean Square(Error) = 4.473E-02.

a. Uses Harmonic Mean Sample Size = 13.115.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = .05.

Figure 5-23: *Statistically Different Boundary Ranks*

Summary

This chapter showed evidence that the boundary ranks assigned according to the SBR are reflected in the prosody of coordinate structure. A recoding of the SBR in terms of a left-to-right implementation (the NSBR) led to a fairly accurate model of the relative strength of boundaries as it is reflected in duration.

The theory proposed here uses a relative notion of boundary rank instead of a categorical one as in the theory of prosodic phonology. The left-to-right implementation of the SBR, the NSBR, provides a model of prosody that is compatible with incremental production. Boundaries are scaled in strength relative to earlier boundaries.

A certain amount of look-ahead adjusts the strengths of the boundary after the first constituent in such a way that the subsequent boundaries can be realized in a reasonable range. Look-ahead requires anticipating the complexity of the upcoming structure. The precise measure of complexity used here was the sum of the upcoming boundary ranks—but whether or not this is the right measure is a question for further research.

The look-ahead in this experiment was fairly extensive, possibly due to the fact that the task was such that it facilitated anticipating the entire structure of the answer. A different kind of task might show less look-ahead and more look-back scaling relative to earlier material.